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Generative artificial intelligence and internationalization green innovation: Roles of supply chain innovations and AI regulation for SMEs

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ABSTRACT

This study investigates how generative artificial intelligence (GenAI) impacts internationalization green innovation performance in cross-border e-commerce small and medium-sized enterprises (SMEs). Drawing on the resource-based view, we employed a mixed-methods approach, including partial least squares structural equation modeling (PLS-SEM), importance-performance map analysis (IPMA), fuzzy-set qualitative comparative analysis (fsQCA), and semi-structured interviews with 377 cross-border e-commerce SMEs. Our findings reveal that GenAI capability positively influences internationalization green innovation performance, both directly and indirectly, through supply chain exploratory and exploitative innovation. Notably, artificial intelligence (AI) regulation strengthens, rather than weakens, the positive relationship between GenAI capability and supply chain innovation. This highlights the potential for responsible AI governance to foster sustainable practices. The study contributes to understanding technology's role in societal transformation by illuminating how GenAI can promote sustainable business practices in internationalization contexts. Our findings offer valuable insights for policymakers and business leaders on effectively managing the societal implications of AI adoption while pursuing sustainability goals.

1. Introduction

The emergence of generative artificial intelligence (GenAI) represents a profound technological shift with far-reaching implications for society's sustainable development (Dabić et al., 2023; Li, Song, et al., 2024; Ochieng et al., 2024; Sarkar et al., 2025; Wang & Zhang, 2024). As a subset of AI capable of creating novel content and solutions, GenAI presents unprecedented opportunities for enterprises to enhance productivity, streamline processes, and gain competitive advantages (Rajaram & Tinguely, 2024). However, the integration of GenAI into business practices also raises complex challenges related to implementation, ethics, and regulation (Wach et al., 2023), particularly as firms face increasing pressure to expand globally while minimizing environmental impact (Dabić et al., 2024). The intersection of GenAI capabilities with international green innovation represents a critical yet underexplored area of research with significant implications for the social, economic, and philosophical transformation of the world.

While studies have examined the broader societal impacts of AI on sustainability (Kulkov et al., 2024) and innovation (Mariani et al., 2023), the specific role of GenAI in driving green innovation performance for internationalizing firms remains unclear, especially for small and medium-sized enterprises (SMEs) that often lack the resources of larger corporations but may benefit significantly from GenAI adoption (Al-Emran & Griffy-Brown, 2023). Moreover, the mediating mechanisms through which GenAI influences green innovation outcomes, such as supply chain processes, have received limited attention (Fosso Wamba et al., 2024), as has the rapidly evolving regulatory landscape surrounding AI, which has potential implications for how firms leverage GenAI for international expansion and sustainability initiatives (Díaz-Rodríguez et al., 2023). Given these complex dynamics, there is a pressing need to develop a more nuanced understanding of how GenAI capabilities can be effectively harnessed to enhance internationalization green innovation performance, particularly for SMEs navigating global markets and environmental pressures.

This study aims to address this gap by examining the mediating roles of supply chain exploratory and exploitative innovation in the relationship between GenAI capability and internationalization green innovation performance within the context of cross-border e-commerce SMEs in China. Drawing on the resource-based view (RBV) and the practice-based view (PBV) proposed by Li, Song, et al. (2024), we

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theorize that GenAI capability enhances both exploratory and exploitative supply chain innovation, which in turn drives green innovation outcomes in international markets. We further posit that the regulatory environment surrounding AI shapes these relationships. By investigating these mechanisms, this study seeks to shed light on the operational pathways through which GenAI can be effectively harnessed to drive both environmental and economic outcomes for internationalizing SMEs, contributing to the global discourse on the role of technology in shaping a more sustainable future.

This research makes several important contributions to the literature at the intersection of technological change and societal transformation. First, it advances the understanding of GenAI and green innovation by developing an integrated framework that explicates the mechanisms through which GenAI capabilities influence internationalization green innovation performance. Second, by investigating the mediating roles of supply chain innovation, this research illuminates the operational pathways through which GenAI enhances green innovation performance, addressing calls for more granular insights into AI's effects on supply chain processes (Fosso Wamba et al., 2024). Third, the examination of AI regulation as a moderator contributes to the nascent literature on AI governance (De Almeida et al., 2021), offering empirical evidence on how regulatory factors shape the relationship between AI capabilities and innovation outcomes.

From a practical standpoint, this study provides actionable insights for SME managers seeking to leverage GenAI for sustainable competitive advantage in global markets and for policymakers striving to balance innovation promotion with responsible AI governance. By delineating the specific mechanisms through which GenAI drives green innovation, the findings can guide firms in strategically deploying AI resources to enhance their environmental performance while navigating the complexities of the international regulatory landscape. Overall, this research advances the scholarly and theoretical understanding of GenAI's role in driving sustainable internationalization while providing a roadmap for practitioners to harness AI's potential in achieving green innovation objectives, thereby contributing to the intentional shaping of technology's impact on society.

The paper proceeds as follows: We first review relevant literature and develop our hypotheses, then describe our methodology and present our findings. We conclude by discussing theoretical and practical implications, limitations, and future research directions.

2. Literature review

GenAI represents a transformative technological force that is fundamentally reshaping societal dynamics and business practices across the globe. With its capacity for content creation and problem-solving, GenAI is not merely a tool for enhancing productivity but a catalyst for profound social and economic transformation (Ooi et al., 2023; Rajaram & Tinguely, 2024). This technological advancement carries significant implications for sustainable development and social responsibility, particularly in how organizations address environmental challenges while navigating complex global markets (Clemente-Almendros et al., 2024). The societal impact of GenAI extends beyond mere operational efficiency, raising important questions about how this technology can be thoughtfully integrated into business practices to serve broader social and environmental goals, especially within the context of small and medium-sized enterprises (SMEs) that form the backbone of many economies.

Recent research by Shore et al. (2024) empirically demonstrates how GenAI can enhance entrepreneurial resilience in SMEs during crises. Their study highlights the role of GenAI in enabling firms to adapt to market turbulence and maintain competitiveness. This is particularly relevant to our study as it provides empirical evidence of GenAI's positive impact on SMEs, aligning with our focus on cross-border e-commerce SMEs. Dubey et al. (2024) provide a theoretical toolbox for integrating GenAI into operations and supply chain management, which is particularly relevant for benchmarking practices. Their work underscores the potential of GenAI to revolutionize businesses and create significant value by enabling more effective and efficient supply chain operations. Li, Song, et al. (2024) adopt a practice-based view to examine generative AI usage's effect on sustainable supply chain performance. They find that two types of sustainable supply chain practices—green supply chain collaboration and circular economy implementation—emerge as serial mediators connecting this relationship. The concept of supply chain survivability, as explored by Sarkar et al. (2025), emphasizes the importance of maintaining vital operations and providing value despite formidable challenges.

The existing literature has increasingly recognized AI's potential to drive sustainable societal development (Bolón-Canedo et al., 2024; Kulkov et al., 2024; Ochieng et al., 2024). For instance, studies have explored AI's role in optimizing resource allocation, improving energy efficiency, and promoting circular economy models (Bolón-Canedo et al., 2024). However, these studies predominantly focus on general AI applications rather than the specific capabilities and implications of GenAI. Moreover, while research has examined AI's broader impact on sustainability (Budhwar et al., 2023; Zhang, 2024) and international business practices (Hasan & Ojala, 2024), a critical gap exists in understanding how GenAI, with its unique content creation and problem-solving capabilities, specifically shapes social and environmental outcomes in the context of internationalizing SMEs. This gap is particularly significant given Technology in Society's emphasis on examining how technological decisions, especially concerning emerging technologies like GenAI, influence societal transformation (Clemente-Almendros et al., 2024). The existing literature lacks a nuanced exploration of the mechanisms through which GenAI impacts organizational practices and environmental performance. While some scholars acknowledge the role of supply chain management in AI-driven sustainability (Fosso Wamba et al., 2024; Ghobakhloo et al., 2024), a more granular understanding is needed of how GenAI shapes organizational behavior and decision-making in pursuit of sustainable practices, particularly in the context of internationalization.

Our study addresses these gaps by investigating the specific role of GenAI in driving green innovation performance for internationalizing SMEs. We focus on the mediating mechanisms of supply chain exploratory and exploitative innovation, providing a more detailed examination of how GenAI shapes organizational practices and environmental outcomes. This approach allows us to move beyond broad discussions of AI's impact and delve into the specific operational pathways through which GenAI influences sustainability in an international business context. Moreover, we incorporate the moderating role of AI regulation, a rapidly evolving area that significantly impacts how firms leverage GenAI for international expansion and sustainability, thus contributing to the nascent literature on responsible AI governance (Díaz-Rodríguez et al., 2023).

Our focus on SMEs is motivated by their significant contribution to global economies and their increasing need to adopt sustainable practices while expanding internationally (Dabić et al., 2024). SMEs often face resource constraints and unique challenges in adopting new technologies compared to larger corporations (Clemente-Almendros et al., 2024). Recent studies highlight GenAI's democratizing potential for SMEs, suggesting that it can level the playing field by providing access to advanced capabilities previously available only to larger firms (Rajaram & Tinguely, 2024). However, questions remain about how this technology can be effectively leveraged by SMEs to advance both societal and environmental objectives in an increasingly globalized world, particularly within the complex interplay of internationalization, sustainability, and emerging technology adoption. We specifically focus on Chinese cross-border e-commerce SMEs for several reasons. First, China is a global leader in both e-commerce and AI development, making it a particularly relevant context for studying the intersection of GenAI, internationalization, and sustainability (Deloitte, 2024; Li et al., 2023). Second, cross-border e-commerce is a rapidly growing sector in China,

with SMEs playing a crucial role in driving its expansion (Li et al., 2024a, b). These firms are at the forefront of leveraging digital technologies to compete in global markets, making them an ideal context for examining the impact of GenAI. Third, the Chinese government has placed a strong emphasis on promoting both green development and the responsible development of AI, creating a unique regulatory environment that shapes how firms adopt and utilize AI technologies (Zhang, 2024). This setting allows for a nuanced examination of how regulatory factors interact with technological capabilities to influence innovation outcomes. By studying Chinese cross-border e-commerce SMEs, we can gain valuable insights into how GenAI can be harnessed to drive both economic growth and environmental sustainability in a dynamic and strategically important sector.

Moreover, the literature has yet to fully explore the social mechanisms through which GenAI influences organizational practices and environmental outcomes. While some scholars acknowledge supply chain management's role in AI-driven sustainability (Fosso Wamba et al., 2024), we need a more nuanced understanding of how GenAI shapes organizational behavior and decision-making in pursuit of sustainable practices. This study addresses this gap by examining how GenAI capability influences supply chain innovation patterns and environmental performance, considering the broader societal implications of these technological choices. This investigation aligns with Technology in Society's focus on understanding how social forces shape technological decisions and their subsequent impact on society. By exploring these relationships within the context of internationalizing SMEs, this research contributes to our understanding of how technological advancement can be harnessed to promote both economic development and environmental sustainability, key concerns in the journal's scope.

3. Theory and hypotheses

3.1. Resource-based view

Our study is grounded in the Resource-Based View (RBV), which posits that firms can achieve sustainable competitive advantage by acquiring, developing, and leveraging valuable, rare, inimitable, and non-substitutable resources (Barney, 1991). This perspective emphasizes the internal capabilities of the firm as the primary source of competitive advantage, suggesting that firms achieve superior performance by leveraging their unique resources and competencies (Wang & Esperança, 2023). The RBV has been widely applied in various strategic management contexts, including innovation (Kumar et al., 2024), internationalization (Zhang & Deborah, 2024), and sustainability (Alkaraan et al., 2024). It provides a robust theoretical lens for understanding how firms can leverage their internal capabilities to achieve strategic goals in dynamic and competitive environments. We acknowledge that RBV has evolved beyond its initial focus on internal resources to encompass the importance of external relationships and networks (Hitt et al., 2016a,b). In internationalization and supply chain management, firms can access valuable resources and knowledge through collaborations with suppliers, customers, and other stakeholders (Tiwari et al., 2024). These external relationships can be particularly crucial for SMEs that may lack certain internal resources (Al-Emran & Griffy-Brown, 2023).

In this study, we define resources as the tangible and intangible assets that a firm owns or controls, including its GenAI technology, data, and skilled personnel (Barney, 1991; Wang & Zhang, 2024). Capabilities, on the other hand, refer to the firm's ability to deploy these resources effectively to achieve desired outcomes, such as improved supply chain efficiency or the development of innovative green products (Sirmon et al., 2007). We argue that it is the dynamic interaction between resources and capabilities that leads to a sustained competitive advantage (Hitt et al., 2016a,b). We argue that GenAI capabilities represent a complex, firm-specific resource encompassing technological infrastructure, specialized human capital, and organizational routines for effectively utilizing GenAI (Mikalef & Gupta, 2021). These capabilities, when effectively deployed, enable firms to optimize supply chain processes, enhance decision-making quality, and foster the development of innovative green solutions that are difficult for competitors to imitate (Fosso Wamba et al., 2024). This is particularly relevant for SMEs, which often face resource constraints but can significantly benefit from the efficiencies and innovative potential offered by GenAI (Al-Emran & Griffy-Brown, 2023). However, we also recognize the limitations of a purely static RBV approach. As Hitt et al. (2016a) point out, resources can depreciate or become obsolete over time. Therefore, we acknowledge the importance of dynamic capabilities in the context of GenAI, aligning with the arguments of Teece et al. (1997). We posit that successful SMEs will not only possess GenAI capabilities but also exhibit the ability to continuously adapt, learn, and reconfigure these capabilities to respond to evolving market demands and technological advancements.

Furthermore, we posit that the integration of GenAI with existing organizational resources and knowledge bases can create synergistic effects, leading to the development of idiosyncratic capabilities that support internationalization green innovation performance (Akhtar et al., 2024). By applying the RBV lens, we aim to elucidate the specific mechanisms through which GenAI capabilities contribute to sustainable competitive advantage in the global marketplace while simultaneously addressing environmental concerns. This theoretical grounding connects directly to our research problem by framing GenAI not just as a technological tool, but as a strategic resource that, when combined with effective supply chain management and appropriate regulatory frameworks, can drive significant improvements in green innovation performance for internationalizing SMEs.

3.2. GenAI capability and internationalization green innovation performance

RBV posits that firms can achieve sustainable competitive advantage by leveraging valuable, rare, inimitable, and non-substitutable resources and capabilities (Barney, 1991). In the context of international green innovation, GenAI capability represents a cutting-edge technological resource that can potentially transform firms' innovation processes and outcomes. GenAI's ability to generate novel ideas, optimize complex systems, and process vast amounts of data makes it a powerful tool for addressing sustainability challenges in global markets (Kulkov et al., 2024). By harnessing GenAI capabilities, firms can enhance their ability to develop eco-friendly products, improve environmental performance of processes, and implement innovative green technologies across international operations (Lapologang & Zhao, 2023). The advanced data processing and predictive capabilities of GenAI can enable firms to identify emerging sustainability trends, optimize resource allocation, and make more informed decisions regarding green innovation initiatives in diverse international contexts (Akhtar et al., 2024). GenAI can facilitate knowledge sharing and collaboration across global value chains, fostering a more holistic approach to sustainability innovation (Fosso Wamba et al., 2024). As firms integrate GenAI into their innovation processes, they can accelerate the development and commercialization of green technologies, leading to improved environmental performance and competitive advantage in international markets (Zhang, 2024). Beyond the initial acquisition of GenAI technologies, firms must invest in ongoing training and development to ensure their workforce can effectively utilize and adapt these technologies (Mikalef & Gupta, 2021). This dynamic process of capability development is crucial for sustaining the competitive advantage derived from GenAI (Teece et al., 1997). Therefore, this study proposes.

H1. GenAI capability positively influences internationalization green innovation performance.

3.3. GenAI capability and supply chain ambidexterity innovation

We further build upon the RBV to argue that GenAI capability, as a valuable and potentially inimitable resource, enhances a firm's capacity for supply chain ambidexterity innovation. GenAI's potential is not static; it evolves with new algorithms and applications (Wang & Zhang, 2025). Firms must continuously monitor these advancements and adapt their supply chain strategies accordingly. This requires a dynamic capability to absorb and integrate new knowledge, ensuring that the firm's GenAI capabilities remain relevant and effective (Hitt et al., 2016a,b). Supply chain ambidexterity refers to a firm's ability to simultaneously pursue both exploratory and exploitative innovation within its supply chain (Nie et al., 2022). This dual focus is crucial for achieving sustained competitive advantage in dynamic environments. Supply chain exploratory innovation involves seeking radically new knowledge, experimenting with novel technologies, and developing entirely new processes and configurations within the supply chain to enhance its flexibility and adaptability (Wael Al-khatib et al., 2024). It focuses on creating fundamentally different approaches to managing the flow of goods, information, and finances across the supply chain network. For instance, exploring the use of blockchain for enhanced traceability and transparency or experimenting with autonomous delivery systems for last-mile logistics would fall under exploratory innovation. Supply chain exploitative innovation refers to the refinement and improvement of existing supply chain processes, leveraging current knowledge and technologies to enhance efficiency, reduce costs, and increase reliability (Lin et al., 2022). It focuses on optimizing existing operations through incremental changes and continuous improvement. Examples include using data analytics to optimize inventory levels, fine-tuning delivery routes for reduced fuel consumption, or implementing advanced forecasting techniques for better demand planning.

The effective deployment of GenAI capabilities in the supply chain requires not only the technology itself (a resource) but also the organizational ability to integrate it into existing processes and leverage it for strategic decision-making (a capability). This capability, developed through the bundling of GenAI technology, data, and skilled personnel, enables firms to achieve both exploratory and exploitative innovation, ultimately contributing to a sustained competitive advantage (Hitt et al., 2016a,b; Lin et al., 2022). GenAI, with its ability to analyze vast datasets, identify patterns, and generate novel solutions, can empower firms to enhance both dimensions of supply chain ambidexterity (Akter et al., 2023). Specifically, GenAI can foster supply chain exploratory innovation by enabling firms to simulate and evaluate a wider range of potential solutions, identify emerging trends, and experiment with novel approaches to supply chain management (Wael Al-khatib et al., 2024). For instance, GenAI can be used to model and analyze the impact of incorporating new technologies or sustainable practices into the supply chain, allowing firms to identify potential risks and opportunities before making significant investments (Ghobakhloo et al., 2024). By facilitating such data-driven exploration, GenAI can help firms break free from traditional supply chain paradigms and uncover innovative solutions that enhance efficiency, resilience, and sustainability (Madzík et al., 2024). This exploration of novel supply chain configurations and strategies is crucial for firms seeking to adapt to rapidly changing market demands and technological advancements, ultimately contributing to enhanced competitiveness and long-term success.

Simultaneously, GenAI can also drive supply chain exploitative innovation by enabling firms to optimize existing processes, improve efficiency, and enhance the reliability of their supply chain operations (Lin et al., 2022). GenAI can analyze real-time data from various sources across the supply chain, identify bottlenecks, and recommend adjustments to streamline operations and reduce costs (Fosso Wamba et al., 2024). This data-driven approach to supply chain optimization can lead to significant improvements in areas such as inventory management, logistics, and production planning (Kumar et al., 2024). For example, GenAI-powered systems can predict demand fluctuations with greater accuracy, enabling firms to optimize inventory levels, reduce waste, and minimize stockouts (Tian et al., 2023). Similarly, GenAI can be used to optimize delivery routes, track shipments in real-time, and proactively identify potential disruptions, leading to more efficient and reliable logistics operations (Ooi et al., 2023). This continuous improvement of existing supply chain processes through GenAI-driven insights is essential for firms to maintain operational excellence, reduce costs, and enhance customer satisfaction (Ghali et al., 2024), all of which contribute to a stronger competitive position in the market. Therefore, we propose that.

H2. GenAI capability positively influences supply chain exploratory innovation.

H3. GenAI capability positively influences supply chain exploitative innovation.

3.4. Supply chain ambidexterity innovation and internationalization green innovation performance

Supply chain ambidexterity innovation, encompassing both exploratory and exploitative dimensions, plays a crucial role in enhancing firms' internationalization green innovation performance (Wael Al-khatib et al., 2024). Supply chain exploratory innovation focuses on developing novel solutions and radical approaches to managing global supply chain operations, which can lead to breakthrough environmental innovations (Tian et al., 2023). By actively seeking unconventional supply chain solutions and experimenting with revolutionary processes, firms can discover new ways to reduce their environmental footprint and create eco-friendly products for international markets (Fosso Wamba et al., 2024). Moreover, exploratory innovation in supply chains can help firms identify and leverage emerging green technologies, potentially leading to disruptive sustainability advancements in their global operations (Akhtar et al., 2024). The RBV suggests that such unique and difficult-to-imitate exploratory capabilities can serve as a source of competitive advantage in pursuing green innovation across international markets (Madzík et al., 2024). Collaborating with external partners, such as suppliers and research institutions, can provide access to specialized knowledge and technologies that enhance a firm's ability to develop and implement green innovations (Tiwari et al., 2024). These external relationships can be particularly valuable for SMEs seeking to overcome resource constraints and compete in international markets (Dabić et al., 2024).

Complementing exploratory efforts, supply chain exploitative innovation focuses on refining and optimizing existing processes to enhance efficiency and reliability in international operations (Wael Al-khatib et al., 2024). By implementing incremental improvements to established supply chain practices across global markets, firms can steadily enhance their environmental performance and resource efficiency (Fosso Wamba et al., 2024). Exploitative innovation enables firms to fine-tune their current green technologies and strategies, maximizing their effectiveness in diverse international contexts (Kumar et al., 2024). From a resource-based perspective, the ability to continuously optimize and adapt existing green supply chain capabilities for different markets can lead to sustained competitive advantage in international green innovation (Ghobakhloo et al., 2024). This consistent refinement of environmentally friendly practices across global supply chains can result in cumulative improvements in firms' overall green innovation performance (Zhang & Deborah, 2024). Consequently, this study proposes.

H4. Supply chain exploratory innovation positively influences internationalization green innovation performance.

H5. Supply chain exploitative innovation positively influences internationalization green innovation performance.

3.5. Supply chain ambidexterity innovation as mediator

Supply chain ambidexterity innovation, encompassing both exploratory and exploitative dimensions, is likely to serve as a crucial mediating mechanism through which GenAI capability influences internationalization green innovation performance. Green supply chain collaboration and circular economy implementation serve as important mediators in this relationship (Li, Song, et al., 2024). This mediating role is rooted in the RBV, which posits that firms can achieve sustainable competitive advantage by leveraging valuable, rare, inimitable, and non-substitutable resources and capabilities (Wang & Esperança, 2023). GenAI capability, as a valuable technological resource (Akter et al., 2023), can enhance firms' supply chain ambidexterity by enabling them to both explore novel solutions and exploit existing processes for enhanced efficiency, responsiveness, and sustainability (Wael Al-khatib et al., 2024). This enhanced supply chain ambidexterity, in turn, can drive green innovation outcomes in international markets by facilitating the development and implementation of eco-friendly products, processes, and technologies (Fosso Wamba et al., 2024; Tian et al., 2023). GenAI can foster supply chain exploratory innovation by enabling firms to simulate and evaluate a broader range of potential solutions (Ghobakhloo et al., 2024), identify emerging trends, and experiment with novel approaches to supply chain management (Madzík et al., 2024). Simultaneously, GenAI can also drive supply chain exploitative innovation by enabling firms to optimize existing processes (Kumar et al., 2024), improve efficiency, and enhance the reliability of their supply chain operations (Lin et al., 2022). This dual capability of GenAI to foster both exploratory and exploitative supply chain innovation is expected to create a synergistic effect, leading to enhanced green innovation performance in international markets (Lapologang & Zhao, 2023). Therefore, this study proposes.

H6. Supply chain exploratory innovation mediates the relationship between GenAI capability and internationalization green innovation performance.

H7. Supply chain exploitative innovation mediates the relationship between GenAI capability and internationalization green innovation performance.

3.6. Artificial intelligence regulation as moderator

While GenAI capability and supply chain ambidexterity innovation can drive internationalization green innovation performance, the regulatory environment surrounding AI may shape these relationships. The RBV suggests that a firm's competitive advantage stems from its unique bundle of valuable and rare resources and capabilities (Kumar et al., 2024). Internal reconfiguration focuses on optimizing and restructuring existing internal resources, while external integration emphasizes acquiring external resources (Lu et al., 2025). However, the value and rarity of AI capabilities may be constrained by regulatory restrictions. Stringent AI regulations can limit firms' ability to leverage their AI resources and capabilities fully, potentially hampering innovation outcomes (De Almeida et al., 2021). Regulatory constraints may increase compliance costs and reduce the flexibility of AI applications, thereby weakening the positive impact of GenAI capabilities on green innovation performance in international markets (Díaz-Rodríguez et al., 2023). Moreover, AI regulations may restrict data access and usage, which are critical for developing and deploying effective AI solutions, thus potentially moderating the relationship between AI capabilities and supply chain innovation (Fosso Wamba et al., 2024). Strict regulatory frameworks may also slow down the pace of AI-driven innovation by imposing additional approval processes and safety requirements, which could particularly affect exploratory innovation efforts (Chowdhury et al., 2024).

The impact of AI regulation on supply chain innovation is multifaceted. While regulations aim to ensure responsible AI use, they may inadvertently hinder the exploratory aspects of supply chain innovation. Exploratory innovation often involves experimenting with novel approaches and pushing technological boundaries (Nie et al., 2022). However, stringent AI regulations may create barriers to such experimentation by imposing strict guidelines on AI development and deployment (Wach et al., 2023). This regulatory environment may constrain firms' ability to fully explore the potential of GenAI in revolutionizing supply chain processes, thereby weakening the positive relationship between AI capabilities and supply chain exploratory innovation (Ghobakhloo et al., 2024). Compliance requirements may divert resources from innovative activities, further dampening the exploratory potential of AI in supply chains (Akhtar et al., 2024).

While AI regulation may impede exploratory innovation, its impact on exploitative innovation in supply chains warrants separate consideration. Exploitative innovation focuses on refining and optimizing existing processes, which aligns more closely with regulatory compliance objectives (Wael Al-khatib et al., 2024). However, even in this context, overly restrictive AI regulations may hinder firms' ability to fully leverage their AI capabilities for incremental improvements in supply chain operations (Ooi et al., 2023). Regulatory constraints may limit the scope of AI applications in process optimization, data analysis, and decision-making, potentially weakening the relationship between GenAI capabilities and supply chain exploitative innovation (Kumar et al., 2024). Moreover, the need to navigate complex regulatory landscapes may slow down the implementation of AI-driven improvements, reducing the efficiency gains that exploitative innovation aims to achieve (Feuerriegel et al., 2024). Therefore, this study proposes.

H8. Artificial intelligence regulation negatively moderates the relationship between GenAI capability and internationalization green innovation performance.

H9. Artificial intelligence regulation negatively moderates the relationship between GenAI capability and supply chain exploratory innovation.

H10. Artificial intelligence regulation negatively moderates the relationship between GenAI capability and supply chain exploitative innovation.

3.7. GenAI and green innovation research model

To account for potential confounding factors, this study includes enterprise size and enterprise age as control variables. These variables are commonly acknowledged as influencing factors in innovation adoption and performance, particularly for SMEs (Rajaram & Tinguely, 2024). Incorporating these controls enhances the robustness of the model and allows for a more precise assessment of the relationships between the core constructs. Fig. 1 presents the GenAI & Green Innovation research model, depicting the hypothesized relationships between GenAI capability, supply chain ambidexterity innovation (exploratory and exploitative), internationalization green innovation performance, and AI regulation, while also accounting for the control variables. The following sections will elaborate on the development of the hypotheses and the rationale behind the proposed relationships.

4. Methods

4.1. Research flowchart

This study employed a multi-stage research process to rigorously examine the relationships between GenAI capability, supply chain ambidexterity innovation, and internationalization green innovation performance, as illustrated in Fig. 2.

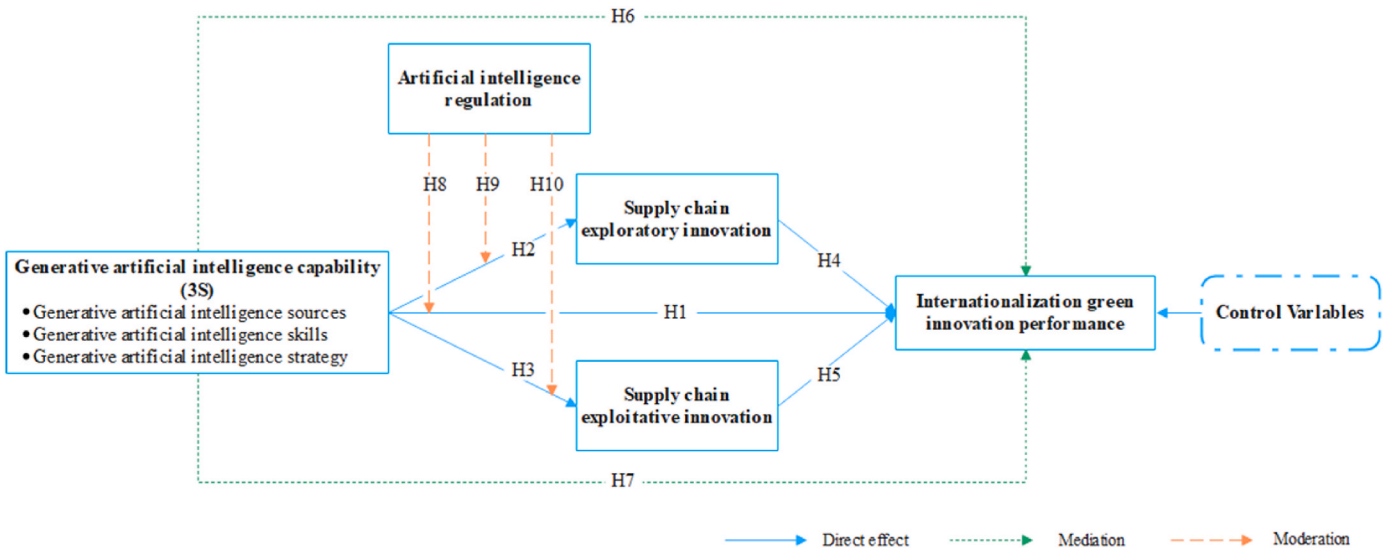


Fig. 1. Research model.

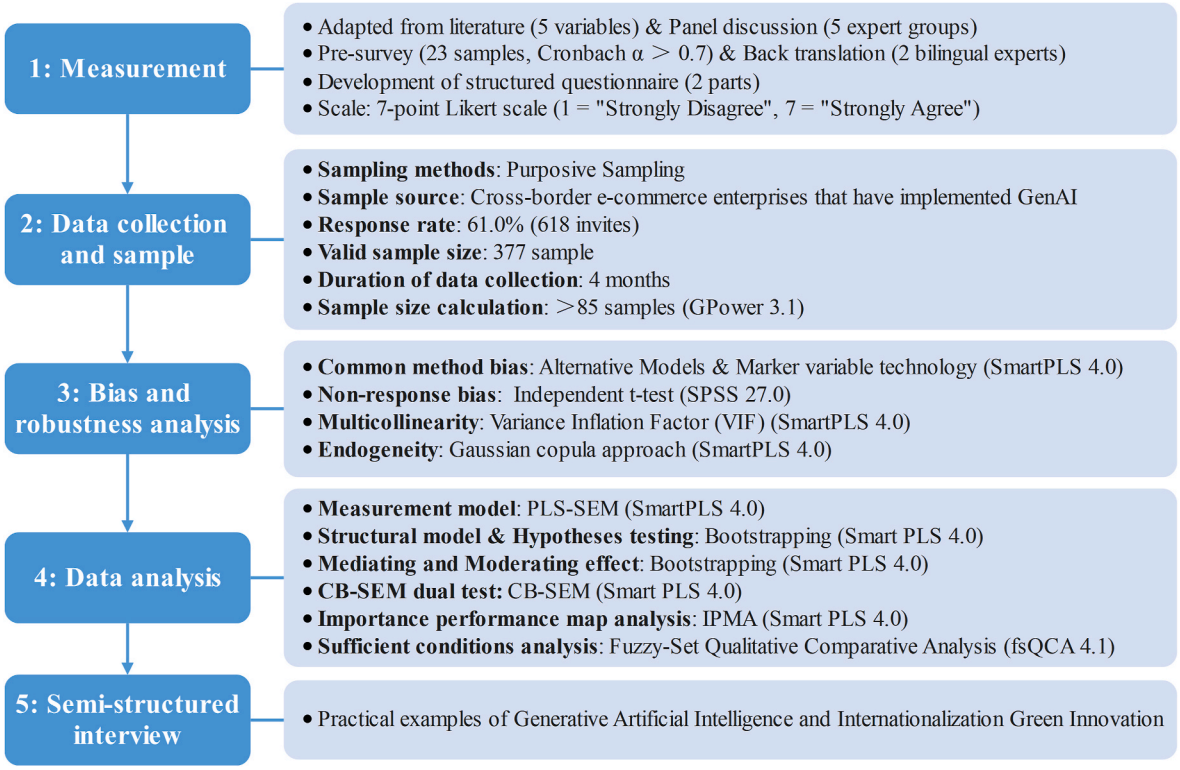


Fig. 2. Research flowchart.

4.2. Research context

This study is situated within the dynamic landscape of international business, where SMEs are increasingly seeking to expand globally while simultaneously navigating the imperative for sustainable practices (Wang & Esperança, 2023). The rapid emergence of GenAI presents both unprecedented opportunities and complex challenges for these firms. GenAI, with its capacity for content creation and problem-solving, offers SMEs novel avenues for enhancing productivity, streamlining processes, and achieving competitive advantage in global markets (Rajaram & Tinguely, 2024). However, the integration of GenAI into business practices also raises considerations related to implementation, ethics, and regulation (Wach et al., 2023), particularly in the context of internationalization and sustainability efforts.

This research specifically focuses on cross-border e-commerce SMEs in China, a sector characterized by intense competition, rapid technological advancements, and growing pressure to adopt sustainable practices. China's e-commerce market is the largest in the world, with cross-border e-commerce playing a significant role in the country's economic growth (Li et al., 2023). These SMEs are at the forefront of leveraging digital technologies to compete in global markets (Chen et al., 2022), making them an ideal context for examining the interplay between GenAI capability, supply chain innovation, and internationalization green innovation performance. The Chinese government's emphasis on

promoting green development and its evolving regulatory framework surrounding AI (Zhang, 2024) further enriches the research context, providing a unique setting to investigate how regulatory factors shape the relationship between AI capabilities and innovation outcomes. This setting allows for a nuanced examination of how GenAI capabilities can be effectively harnessed to enhance internationalization green innovation performance, particularly for SMEs navigating global markets and environmental pressures. The Chinese government's dual emphasis on promoting green development and responsible AI creates a unique regulatory environment that shapes how firms adopt and utilize AI technologies (Zhang, 2024). This context provides a valuable opportunity to examine how regulatory factors interact with technological capabilities to influence innovation outcomes, a dimension often overlooked in traditional RBV research (Tiwari et al., 2024).

4.3. Questionnaire design

The development of the measurement instrument followed a rigorous multi-stage process, as recommended by Hulland et al. (2018), to ensure validity and reliability (Jayasena et al., 2024).

Stage 1. Construct Definition and Item Generation: Initially, constructs were defined based on an extensive review of extant literature on GenAI, supply chain innovation, and green innovation. We drew upon established scales and adapted items to fit the specific context of GenAI and green innovation in cross-border e-commerce SMEs (Wael AL-khatib, 2023). Special attention was paid to conceptualizing GenAI capability as a multifaceted construct encompassing technological resources, data availability, financial investment, employee skills, and strategic alignment, in line with recommendations by Mikalef and Gupta (2021).

Stage 2. Expert Panel Review: Following the initial item generation, we convened a panel of five domain experts, including three academics specializing in technology management and two industry professionals with experience in implementing AI solutions in cross-border e-commerce. This panel evaluated the content validity of our instrument through three rounds of iterative feedback, as suggested by Wang and Esperança (2023). Specifically, they assessed each item for relevance to the construct definition, clarity of wording, and comprehensiveness of the construct's domain. This process resulted in the refinement of item wording, the addition of two items to the GenAI capability construct to better capture its strategic dimension, and the elimination of one redundant item in the supply chain exploitative innovation construct. For instance, an item initially phrased as "Our enterprise has sufficient technical resources to implement GenAI" was refined to "Our enterprise has sufficient technical resources (e.g., hardware and software) to implement GenAI" for greater clarity.

Stage 3. Pilot Test and Psychometric Evaluation: We then conducted a pilot test with 23 senior executives from cross-border e-commerce SMEs to assess the instrument's psychometric properties. This sample size aligns with recommendations for pilot testing in scale development (Fosso Wamba et al., 2024). The results demonstrated satisfactory internal consistency reliability, with Cronbach's alpha coefficients exceeding the 0.7 threshold for all constructs. Based on feedback from pilot participants, we further modified item wording to improve clarity and readability. For example, one item in the AI regulation construct was revised from "Legal restrictions influence our decision-making regarding AI" to "Legal restrictions influence our decision-making process regarding AI implementation" to enhance specificity. This iterative process of expert review and pilot testing ensured that our measures were both conceptually sound and empirically robust. Hulland et al. (2018) emphasize the importance of pretests in the form of qualitative interviews or small-scale quantitative studies, we followed this recommendation by conducting our pilot test with senior executives, providing valuable feedback for instrument refinement.

Stage 4. Cross-Linguistic Equivalence: To ensure cross-linguistic equivalence, we implemented a rigorous back-translation procedure, as recommended by Chen et al. (2022). Two bilingual experts, proficient in both English and Chinese, independently translated the questionnaire. Discrepancies were resolved through discussion, with careful attention paid to maintaining conceptual equivalence across cultural contexts.

Stage 5. Final Instrument: The final instrument employed a 7-point Likert scale (1 = "Strongly Disagree" to 7 = "Strongly Agree") to measure key constructs. This scale format was chosen to capture nuanced variations in respondents' perceptions and to align with established practices in the field (Neiroukh et al., 2024).

These rigorous steps in construct operationalization and pretesting, guided by established best practices and the insights from Hulland et al. (2018) and other cited literature, ensure the validity and reliability of our measurement instrument.

4.4. Data collection and sample

This study employed a purposive sampling technique to collect data from cross-border e-commerce SMEs in China that have implemented GenAI technologies (see Table 1). Purposive sampling was chosen to identify information-rich cases with in-depth knowledge of the research topic (Wang & Esperança, 2023). The sampling frame was developed by contacting cross-border e-commerce associations in eastern, northern, western, and southern regions of China to obtain an initial list of potential respondents. This approach ensured geographical diversity and representativeness of the sample. A two-wave time-lagged survey design was utilized to mitigate common method bias. The first wave collected data on firm characteristics, GenAI capability, and supply chain ambidexterity innovation. Respondents were asked to provide their email addresses for the second wave, which was conducted approximately one month later and focused on AI regulation and internationalization green innovation performance. To ensure only knowledgeable respondents participated, screening questions were included at the beginning of each survey wave. The first wave asked, "Has your enterprise implemented GenAI?" while the second wave inquired, "Are you familiar with GenAI and its applications?". Only affirmative responses were allowed to proceed. After data cleaning to remove incomplete responses and those with suspicious answer patterns, a final sample of 377 valid responses was obtained, representing a response rate of 61.0 %. This sample size exceeds the minimum required based on power analysis using G*Power 3.1 software and the "10 times rule" commonly applied in PLS-SEM studies (Hair et al., 2024). Table 2 presents the characteristics of the sample. Over half (55.4 %) held senior management positions, and most had at least a bachelor's degree (84.6 %). This profile suggests the respondents possessed sufficient knowledge to provide reliable information on their firms' AI initiatives and performance.

4.5. Analytical technology

The study employs a sophisticated analytical framework to ensure rigorous examination of the proposed hypotheses and model fit (Vaithilingam et al., 2024). PLS-SEM was selected for its robustness in handling complex models with multiple constructs, which aligns well with the study's objective of exploring the intricate relationships between GenAI capabilities and internationalization green innovation performance (Hair et al., 2024). PLS-SEM is particularly advantageous in exploratory research contexts where theoretical development is ongoing, as it does not require strict assumptions about data distribution and can handle small to medium sample sizes effectively (Fosso Wamba et al., 2024). To provide a more detailed understanding of the constructs' importance and performance, we have incorporated Importance-Performance Map Analysis (IPMA). IPMA extends the standard PLS-SEM results by considering both the importance (total

Table 1
Questionnaire.

Questions		References
Generative artificial intelligence capability (GAIC)	Our enterprise has sufficient technical resources (e.g., hardware and software) to implement GenAI. Our enterprise has access to the necessary data for running GenAI applications. Our enterprise has allocated adequate funding for GenAI projects. Our employees understand the potential applications of GenAI in our business. Our employees have the skills to use GenAI technologies effectively. Our enterprise provides training opportunities for employees to enhance their GenAI skills. Our enterprise has a clear strategy for integrating GenAI into our business processes. Our enterprise actively seeks opportunities to leverage GenAI for innovation. Our enterprise is committed to using GenAI to improve our international competitiveness.	Chen et al., 2022; Mikalef & Gupta, 2021; Neiroukh et al., 2024
Supply chain exploratory innovation (SCRI)	Our enterprise actively seeks novel supply chain solutions that deviate from existing industry norms. We regularly experiment with radical new approaches to manage our international supply chain operations. Our enterprise invests heavily in discovering breakthrough supply chain technologies for our global operations. We frequently introduce revolutionary supply chain processes that challenge traditional methods in our industry. Our enterprise encourages supply chain partners to propose innovative ideas that could transform our international operations.	Nie et al., 2022; Wael Al-khatib et al., 2024
Supply chain exploitative innovation (SCII)	Our enterprise focuses on refining existing supply chain processes to enhance efficiency in our international operations. We regularly implement incremental improvements to our established supply chain practices across global markets. Our enterprise emphasizes the optimization of current supply chain technologies to serve our international customers better. We consistently seek ways to enhance the reliability of our existing supply chain operations in various countries. Our enterprise prioritizes fine-tuning our current supply chain strategies to maximize performance in international markets.	
Artificial intelligence regulation (AIR)	Our enterprise is required to comply with regulations governing the use of AI technologies. There are legal guidelines that limit the scope of our AI applications. Legal restrictions influence our decision-making process regarding AI implementation. There are data protection rules that restrict how we can use AI to process customer information.	De Almeida et al., 2021; Díaz-Rodríguez et al., 2023
Internationalization green innovation performance (IGIP)	Our enterprise has successfully introduced environmentally friendly products/services in international markets. We have significantly improved the environmental performance of our processes for international operations. Our organization has implemented innovative green technologies. We have achieved notable reductions in resource consumption across our international supply chain. Our enterprise has developed eco-friendly product designs that appeal to environmentally.	Lapologang & Zhao, 2023; Marco-Lajara et al., 2023

Table 2
Characteristics.

Characteristics		%
Enterprise size	<100	31.3
	100–200	41.9
	201–300	26.8
Enterprise age	<5	21.8
	5–10	48.8
	≥11	29.4
Administration role	Owner/CEO/COO	55.4
	Managers/Directors	44.6
Gender	Male	64.2
	Female	35.8
Education	Diploma or less	15.4
	BA degree	54.9
	Postgraduate degree	29.7

effects) and performance (average latent variable scores) of constructs in explaining the target construct (Hair et al., 2024). Specifically, IPMA contrasts the total effects of the constructs in the structural model with their average latent variable scores, which are rescaled to a range from 0 to 100. This allows for the identification of constructs that have a relatively high importance in explaining the target construct but exhibit relatively low performance. These constructs are prime candidates for

managerial attention and improvement efforts (Hair et al., 2024). In this study, we use IPMA to identify key areas where SMEs can focus their efforts to enhance internationalization green innovation performance. The results of the IPMA are presented in Fig. 6 and discussed in Section 5.6. Furthermore, we have integrated Fuzzy-Set Qualitative Comparative Analysis (fsQCA) to explore causal complexity and identify multiple configurational paths leading to high levels of internationalization green innovation performance (Wang & Esperança, 2023). fsQCA is a set-theoretic method that allows for the examination of complex causal relationships by identifying configurations of conditions that are sufficient for an outcome to occur. Unlike traditional regression-based methods, fsQCA does not assume linear relationships or that the same causal factors are equally relevant across all cases. Instead, it focuses on identifying combinations of conditions that are consistently associated with the outcome, even if individual conditions are not sufficient on their own. This approach accommodates the non-linear and asymmetric relationships often present in social science research (Wang & Esperança, 2023). In this study, we use fsQCA to explore the different combinations of GenAI capability, supply chain innovation, and AI regulation that lead to high levels of internationalization green innovation performance. The fsQCA analysis process and results are detailed in Section 6.

The combination of PLS-SEM, IPMA, and fsQCA ensures a comprehensive understanding of the data and enhances the validity of the findings. While PLS-SEM provides insights into the relationships between constructs and their relative importance, IPMA offers a more nuanced perspective on the performance of each construct, and fsQCA allows for the identification of complex causal configurations. This multi-method approach strengthens the robustness of our findings and provides a more holistic understanding of the factors driving internationalization green innovation performance in the context of GenAI adoption.

4.6. Common method bias

We are aware of the potential for common method bias (CMB) to inflate observed relationships between variables when data are collected through self-report surveys (Podsakoff et al., 2003, 2024). To address this concern, we employed both procedural and statistical remedies, as recommended by Podsakoff et al. (2024) and Hulland et al. (2018).

We carefully designed the questionnaire to minimize ambiguity and reduce the likelihood of respondents using a single cognitive anchor to answer all questions. This included using clear and concise language, avoiding double-barreled questions, and varying the scale anchors and formats across different constructs (Podsakoff et al., 2003). As Hulland et al. (2018) point out that using different response scales for different constructs is often not sufficient, we further employed the following remedies. We assured respondents of the confidentiality and anonymity of their responses to reduce evaluation apprehension and social desirability bias (Podsakoff et al., 2003). We introduced a time lag of approximately one month between the collection of data on predictor and criterion variables. This temporal separation helps to reduce the salience of previous responses and minimizes the likelihood of respondents consciously or unconsciously attempting to maintain consistency across their answers (Podsakoff et al., 2024).

Following the recommendations of Podsakoff et al. (2024), we employed a marker variable technique using a theoretically unrelated variable (Perceived Ease of Use, PEU) to assess the potential impact of CMB. The correlations between PEU and the main study variables were found to be non-significant ($p > 0.05$), suggesting that CMB is unlikely to significantly influence our findings. We acknowledge that the marker variable technique has limitations, as noted by Podsakoff et al. (2024), but we believe it provides a valuable additional check for CMB in conjunction with our other procedural and statistical remedies. Alternative Models: We also employed a Confirmatory Factor Analysis (CFA) approach to test for the potential influence of a common method factor. We compared a single-factor model (where all items loaded onto a single common method factor) to our hypothesized measurement model. The results showed that the single-factor model had a significantly poorer fit to the data, indicating that a single common method factor does not account for the majority of the variance in our measures. This is consistent with the recommendations of Podsakoff et al. (2024) for examining alternative models to assess the potential impact of CMB. In line with the strong recommendations of Hulland et al. (2018) and Podsakoff et al. (2024), we have discontinued the use of Harman's single-factor test, recognizing its severe limitations in detecting and controlling for common method bias.

We acknowledge that no single method can completely eliminate the risk of CMB. However, by employing a combination of procedural and statistical remedies, we have taken multiple steps to minimize its potential impact on our findings. The results of our analyses suggest that CMB is unlikely to be a major threat to the validity of our study.

4.7. Non-response bias

To address potential non-response bias, we followed the recommendations of Hulland et al. (2018) and conducted two types of analyses (Wang & Esperança, 2023). First, we compared early and late respondents on key demographic variables (enterprise size and age) using independent samples t-tests. The results showed no significant differences between the two groups ($p = 0.833$ for enterprise size, $p = 0.734$ for enterprise age). Second, we performed Pearson Chi-square tests to compare early and late respondents on categorical variables. Again, no significant differences were found. These analyses suggest that non-response bias is unlikely to be a major concern in our study. As Hulland et al. (2018) suggest, we conducted more than a single test to assess non-response bias and did not rely solely on Armstrong and Overton's (1977) approach.

4.8. Multicollinearity

Multicollinearity affects the stability of regression models, potentially skewing parameter estimates (Hair et al., 2024). This was assessed using Variance Inflation Factor (VIF) calculations. The highest VIF was 3.013, the lowest was 1.722, all below the commonly accepted threshold of 5, indicating no serious multicollinearity issues.

4.9. Endogeneity

The Gaussian Copula method allows researchers and practitioners to detect and correct endogeneity in PLS-SEM. This study utilized the Gaussian copula approach provided by SmartPLS 4 through bootstrapping (5000 samples) to address potential endogeneity. The endogeneity test assessment table showed no Gaussian cofunctions were significant, with p-values equal to or greater than 0.05, indicating no evidence of endogeneity in the model, affirming its robustness (Vaithilingam et al., 2024).

5. Study 1: SEM and IPMA results

To rigorously examine the relationships between GenAI capability, supply chain ambidexterity innovation, and internationalization green innovation performance, this study employs a multi-stage research process (Fig. 2) and a sophisticated analytical framework. This framework integrates four complementary methodologies: PLS-SEM, IPMA, fsQCA, and semi-structured interviews. Each method plays a distinct yet interconnected role in developing the study's integrated framework and addressing the research problem.

PLS-SEM serves as the primary quantitative method to test the hypothesized relationships outlined in our research model (Fig. 1). It allows us to assess the direct and indirect effects of GenAI capability on internationalization green innovation performance, as well as the

mediating roles of supply chain exploratory and exploitative innovation. Furthermore, PLS-SEM enables us to examine the moderating effect of artificial intelligence regulation on these relationships. IPMA complements PLS-SEM by providing a nuanced understanding of the relative importance and performance of each construct in driving internationalization green innovation performance. This analysis helps identify key areas where managerial attention should be focused to maximize the positive impact of GenAI adoption. fsQCA extends the analysis beyond the linear and symmetrical relationships examined by PLS-SEM. It allows us to explore the complex interplay of factors that lead to high levels of internationalization green innovation performance, identifying multiple configurational pathways that involve different combinations of GenAI capability, supply chain innovation, and regulatory context. This method is particularly useful in uncovering causal complexity and equifinality, where different paths can lead to the same outcome. Semi-structured interviews with senior executives from cross-border e-commerce SMEs provide qualitative insights that enrich and contextualize the quantitative findings. These interviews delve into the motivations, challenges, and experiences of firms implementing GenAI for internationalization and sustainability, offering a deeper understanding of the phenomenon under investigation.

The combination of these four methodologies ensures a comprehensive understanding of the data and enhances the validity of the findings. Together, they form the basis of our integrated framework, which elucidates how GenAI capabilities can be effectively harnessed to enhance internationalization green innovation performance, particularly for SMEs navigating global markets and environmental pressures, while also considering the influence of the regulatory environment. Section 5 presents the results of the PLS-SEM and IPMA analyses, while Section 6 details the findings from the fsQCA and the qualitative interviews.

5.1. Model assessment

The assessment of the measurement model reveals robust reliability and validity across all constructs, affirming the soundness of the research framework (Hair et al., 2024). As indicated in Table 3, all item loadings exceed the recommended threshold of 0.7, with Variance Inflation Factor (VIF) values comfortably below 5, suggesting no multicollinearity issues. The Cronbach's alpha, composite reliability (rho_c), and average variance extracted (AVE) for each construct surpass the accepted benchmarks of 0.7 and 0.5, respectively, indicating excellent internal consistency and convergent validity. Notably, the GenAI capability construct exhibits a particularly high Cronbach's alpha of 0.937, underscoring its measurement precision. Discriminant validity, as demonstrated in Table 4, is confirmed through the Fornell-Larcker criterion, where the square root of AVE for each construct exceeds the inter-construct correlations, thus ensuring that constructs are distinct from one another. The Heterotrait-Monotrait Ratio (HTMT) analysis further corroborates these findings, with all values falling within acceptable limits, reinforcing the discriminant validity of the model (Hair et al., 2024).

5.2. Model and hypothesis testing

The results of the PLS-SEM analysis presented in Fig. 3 and Table 5 reveal significant insights into the relationships articulated in the proposed model (Hair et al., 2024). Hypothesis 1, which posits that GenAI capability positively influences internationalization green innovation performance, is supported, albeit with a modest effect size ($\beta = 0.100$, $p < 0.05$, $f^2 = 0.012$). This suggests that while GenAI capability is indeed a driver of green innovation in international contexts, its direct impact is relatively limited, emphasizing the necessity for further exploration of mediating pathways. Hypotheses 2 and 3, concerning the influence of GenAI capability on supply chain exploratory and exploitative innovation, are strongly supported ($\beta = 0.350$, $p < 0.001$, $f^2 = 0.168$; $\beta = 0.390$, $p < 0.001$, $f^2 = 0.213$, respectively). These findings underscore the pivotal role of AI in enhancing both the exploratory and exploitative dimensions of supply chain innovation, highlighting the dual capacity of GenAI to foster adaptability and efficiency within supply chains.

5.3. Mediating and moderating effect

The mediating role of supply chain innovation is further evidenced

Table 3
Reliability and validity.

Construct	Items	Loadings >0.7	VIF <3	Cronbach's Alpha>0.7	rho_a >0.7	rho_c >0.7	AVE >0.5	Mean	SD
Generative artificial intelligence capability (GAIC)	GenAIC1	0.809	2.409	0.937	0.939	0.947	0.666	4.822	1.502
	GenAIC2	0.834	2.659						
	GenAIC3	0.833	2.618						
	GenAIC4	0.812	2.426						
	GenAIC5	0.818	2.432						
	GenAIC6	0.784	2.155						
	GenAIC7	0.807	2.422						
	GenAIC8	0.823	2.442						
	GenAIC9	0.824	2.488						
Supply chain exploratory innovation (SCRI)	SCRI1	0.819	1.956	0.875	0.877	0.909	0.666	4.882	1.411
	SCRI2	0.809	1.896						
	SCRI3	0.834	2.187						
	SCRI4	0.811	1.994						
	SCRI5	0.806	1.996						
Supply chain exploitative innovation (SCII)	SCII1	0.859	2.360	0.899	0.903	0.926	0.713	4.735	1.619
	SCII2	0.808	2.113						
	SCII3	0.820	2.095						
	SCII4	0.884	3.013						
	SCII5	0.850	2.383						
Artificial intelligence regulation (AIR)	AIR1	0.827	1.946	0.847	0.851	0.897	0.685	4.493	1.608
	AIR2	0.826	1.772						
	AIR3	0.845	1.990						
	AIR4	0.811	1.861						
Internationalization green innovation performance (IGIP)	IGIP1	0.807	1.935	0.880	0.880	0.912	0.675	4.757	1.465
	IGIP2	0.832	2.104						
	IGIP3	0.815	2.013						
	IGIP4	0.827	2.136						
	IGIP5	0.828	2.084						

Table 4
Discriminant validity.

Construct	AIR	GenAIC	IGIP	SCII	SCRI
AIR	**0.827**	0.183	0.335	0.330	0.301
GenAIC	0.164	**0.816**	0.383	0.458	0.413
IGIP	0.291	0.349	**0.822**	0.586	0.606
SCII	0.291	0.424	0.524	**0.845**	0.672
SCRI	0.262	0.379	0.533	0.597	**0.816**
Items	**AIR**	**GenAIC**	**IGIP**	**SCII**	**SCRI**
AIR1	**0.827**	0.133	0.221	0.220	0.210
AIR2	**0.826**	0.116	0.275	0.237	0.241
AIR3	**0.845**	0.169	0.238	0.265	0.243
AIR4	**0.811**	0.123	0.224	0.240	0.166
GenAIC1	0.101	**0.809**	0.254	0.334	0.292
GenAIC2	0.144	**0.834**	0.299	0.326	0.336
GenAIC3	0.172	**0.833**	0.275	0.374	0.297
GenAIC4	0.108	**0.812**	0.270	0.306	0.301
GenAIC5	0.182	**0.818**	0.301	0.379	0.325
GenAIC6	0.111	**0.784**	0.253	0.304	0.272
GenAIC7	0.154	**0.807**	0.299	0.324	0.260
GenAIC8	0.139	**0.823**	0.309	0.384	0.330
GenAIC9	0.085	**0.824**	0.294	0.372	0.354
IGIP1	0.213	0.258	**0.807**	0.429	0.434
IGIP2	0.240	0.292	**0.832**	0.438	0.479
IGIP3	0.236	0.288	**0.815**	0.435	0.420
IGIP4	0.268	0.307	**0.827**	0.390	0.406
IGIP5	0.240	0.288	**0.828**	0.457	0.449
SCII1	0.295	0.391	0.485	**0.859**	0.523
SCII2	0.184	0.297	0.435	**0.808**	0.506
SCII3	0.241	0.407	0.369	**0.820**	0.478
SCII4	0.226	0.321	0.448	**0.884**	0.526
SCII5	0.271	0.369	0.466	**0.850**	0.488
SCRI1	0.220	0.364	0.454	0.495	**0.819**
SCRI2	0.240	0.319	0.458	0.521	**0.809**
SCRI3	0.190	0.346	0.413	0.481	**0.834**
SCRI4	0.265	0.271	0.397	0.460	**0.811**
SCRI5	0.150	0.232	0.451	0.473	**0.806**

Note: Bold numbers are AVE square roots. Fornell-Larcker Criterion figures appear below the diagonal in the lower left corner, while matching places show HTMT Ratio analysis.
Note: Bold numbers indicate construct element loading.

through the support for Hypotheses 4 and 5, where both exploratory and exploitative innovations significantly enhance internationalization green innovation performance ($\beta = 0.305$, $p < 0.001$, $f^2 = 0.086$; $\beta = 0.263$, $p < 0.001$, $f^2 = 0.062$, respectively) (Hair et al., 2024). These results affirm the importance of a balanced approach in supply chain strategy that leverages both exploratory and exploitative innovations to drive sustainable outcomes. The mediation effects posited in Hypotheses 6 and 7 are confirmed (Table 6), indicating that supply chain innovation indeed mediates the relationship between GenAI capability and internationalization green innovation performance. These pathways elucidate the indirect mechanisms through which GenAI capability can be leveraged to achieve broader sustainability goals in international markets.

Contrary to the initial hypotheses, the moderating effects of artificial intelligence regulation, as tested in Hypotheses 8, 9, and 10, suggest a nuanced impact (Figs. 4 and 5). While AI regulation does not significantly moderate the relationship between GenAI capability and internationalization green innovation performance ($\beta = 0.009$, $p > 0.05$), it significantly enhances the impact on supply chain exploratory and exploitative innovation ($\beta = 0.319$, $p < 0.001$; $\beta = 0.266$, $p < 0.001$, respectively). These findings imply that regulatory environments may not impede but rather facilitate certain dimensions of supply chain innovation, warranting a reconsideration of regulatory strategies to support innovation.

5.4. PLS-SEM model fit, R^2, Q^2, f^2

The analysis of the PLS-SEM model fit indicators demonstrates the robustness and reliability of the proposed research model (Hair et al.,

Table 5
Hypothesis test.

Hypotheses		β	SD	T	p	f^2	Support?
H1	GAIC → IGIP	0.100	0.049	2.054	0.040	0.012	✓
H2	GAIC → SCRI	0.350	0.046	7.538	0.000	0.168	✓
H3	GAIC → SCII	0.390	0.042	9.205	0.000	0.213	✓
H4	SCRI → IGIP	0.305	0.054	5.620	0.000	0.086	✓
H5	SCII → IGIP	0.263	0.054	4.871	0.000	0.062	✓
H6	GAIC → SCRI → IGIP	–	–	–	–	–	✓
H7	GAIC → SCII → IGIP	–	–	–	–	–	✓
H8	AIR x GAIC → IGIP	0.009	0.045	0.203	0.839	0.000	×
H9	AIR x GAIC → SCRI	0.319	0.039	8.129	0.000	0.147	×
H10	AIR x GAIC → SCII	0.266	0.042	6.399	0.000	0.104	×

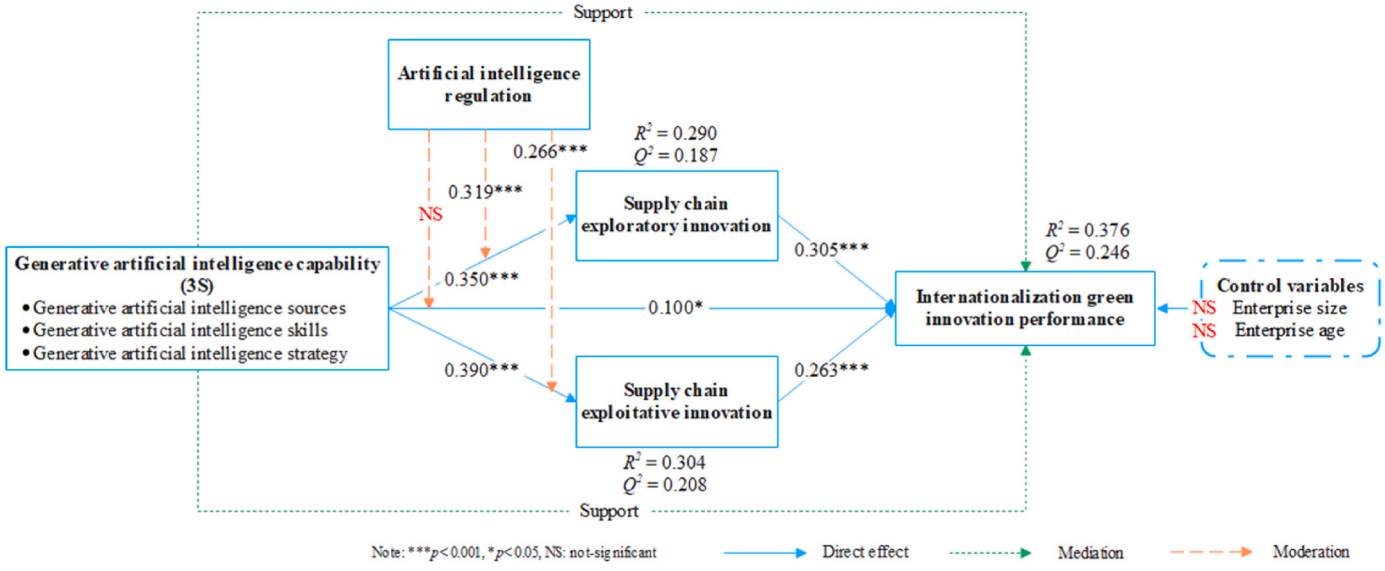


Fig. 3. PLS-SEM model.

Table 6
Mediating and moderating effect.

Effect		β	SD	T	P	CI
Mediating effect	GenAIC → SCRI → IGIP	0.107	0.023	4.569	0.000	[0.065,0.158]
	GenAIC → SCII → IGIP	0.103	0.024	4.278	0.000	[0.059,0.153]
Moderating effect	AIR x GenAIC → IGIP	0.009	0.045	0.203	0.839	[-0.082,0.097]
	AIR x GenAIC → SCII	0.266	0.042	6.399	0.000	[0.183,0.345]
	AIR x GenAIC → SCRI	0.319	0.039	8.129	0.000	[0.238,0.395]

2024). The Standardized Root Mean Square Residual (SRMR) for the saturated model stands at 0.04, indicating an excellent fit between the hypothesized model and the observed data. Complementary to this, the chi-square value of 936.49, despite its sensitivity to sample size, provides further confirmation of the model's adequacy. The Normed Fit Index (NFI) of 0.872 approaches the threshold for good fit, highlighting the model's overall coherence. The d_ULS and d_G values, 1.069 and 0.429 respectively, further underscore the model's structural integrity.

Evaluating the effect size f^2, it becomes evident that GenAI capability significantly influences supply chain exploratory innovation and supply chain exploitative innovation with f^2 values of 0.168 and 0.213, respectively. These substantial effect sizes indicate the pivotal role of GenAIC in driving supply chain ambidexterity within the studied context. The interaction term, AIR x GenAIC, also presents noteworthy effects on SCII and SCRI with f^2 values of 0.104 and 0.147, respectively, suggesting a moderated influence of AI regulation on these relationships.

Cross-validated redundancy measures, expressed as Q^2, provide further insights into the model's predictive relevance. The Q^2 values for Internationalization Green Innovation Performance, SCII, and SCRI are 0.246, 0.208, and 0.187, respectively, all exceeding the threshold of 0, thereby confirming the model's capability to predict these endogenous constructs effectively. This predictive power reinforces the theoretical propositions and empirical findings, offering a comprehensive understanding of the mechanisms through which GenAI capabilities and regulatory factors intersect to influence green innovation outcomes.

5.5. CB-SEM dual test

The CB-SEM dual test offers a robust evaluation of the model's fit (Table 7), confirming its validity and reliability in the context of GenAI and internationalization green innovation (Wael Al-khatib et al., 2024). The ChiSqr/df ratio of 1.734, being well below the threshold of 3, indicates an excellent model fit, suggesting that the proposed model accurately represents the data structure. The RMSEA value of 0.044 further supports this conclusion, demonstrating a good fit and indicating minimal model misspecification. The Goodness of Fit Index (GFI) and the Normed Fit Index (NFI) scores of 0.917 and 0.927, respectively, surpass the critical value of 0.90, underscoring the model's overall adequacy. Although the Adjusted Goodness of Fit Index (AGFI) is slightly below the conventional threshold at 0.899, it remains within acceptable limits, affirming the model's structural integrity. The Parsimony Goodness of Fit Index (PGFI) at 0.755 reflects the model's efficiency in parameter utilization without compromising fit quality. Both the Tucker-Lewis Index (TLI) and the Comparative Fit Index (CFI) exceed the benchmark of 0.90, with values of 0.964 and 0.968, respectively, further validating the model's theoretical robustness. The Standardized Root Mean Square Residual (SRMR) at 0.097, while slightly above the ideal threshold, remains within the bounds of acceptability, indicating the model's residuals are adequately minimized. Collectively, these indices provide strong evidence of the model's fit, supporting its application in exploring the nuanced relationships between GenAI capabilities and green innovation performance in an international context.

5.6. Importance-performance map analysis

IPMA offers a nuanced perspective on the strategic priorities for

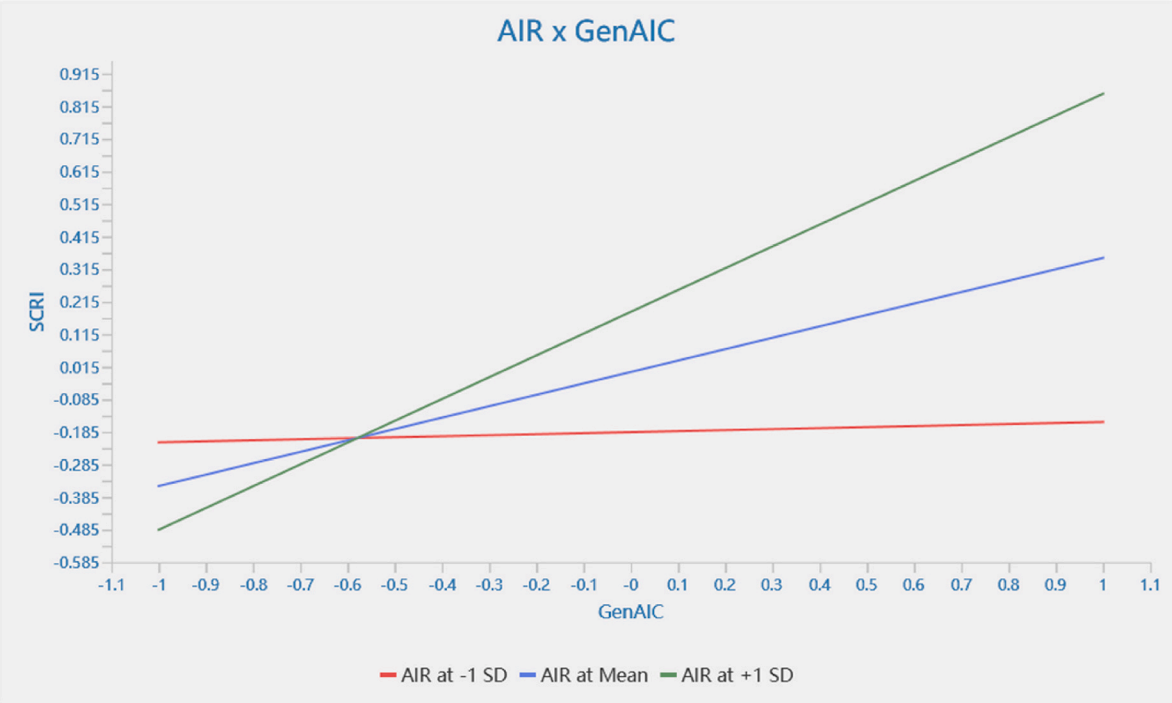


Fig. 4. Moderating effect of artificial intelligence regulation on GenAI capability and supply chain exploratory innovation.

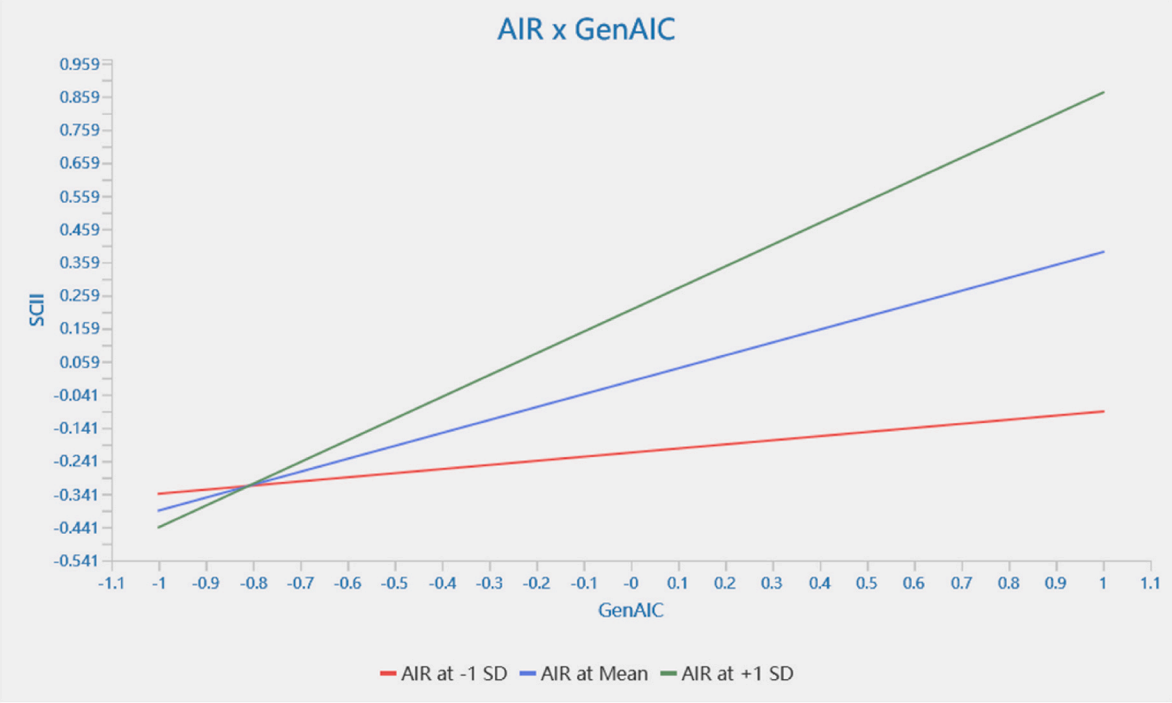


Fig. 5. Moderating effect of artificial intelligence regulation on GenAI capability and supply chain exploitative innovation.

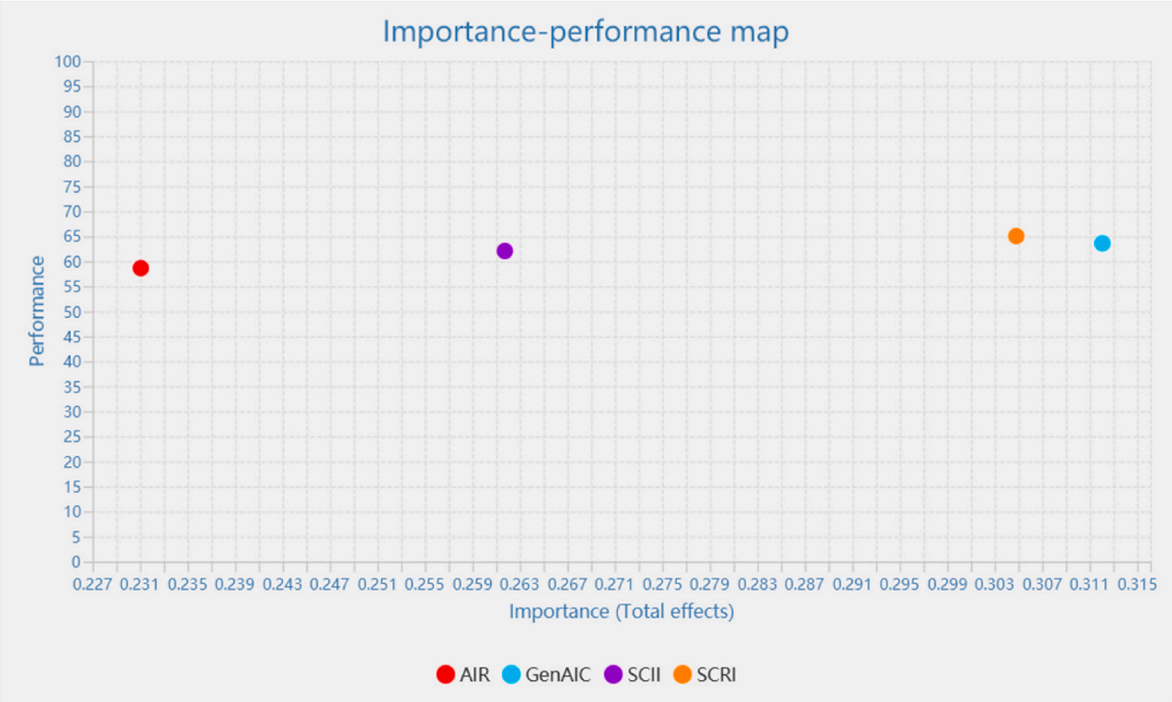


Fig. 6. IPMA.

enhancing internationalization green innovation performance (Hair et al., 2024). Fig. 6 highlights that GenAI capability emerges as the most critical factor, with a total effect of 0.312 and a performance score of 63.605, underscoring its pivotal role in driving sustainable competitive advantage. This is closely followed by supply chain exploratory innovation, which exhibits a total effect of 0.305 and the highest performance score of 65.048, indicating its substantial impact and current effectiveness in fostering green innovation. Supply chain exploitative innovation also demonstrates a significant influence with a total effect of 0.261 and a performance score of 62.080, reflecting its essential contribution to optimization and efficiency in international operations. Artificial intelligence regulation, while having the lowest total effect of 0.231, presents a notable performance score of 58.653, suggesting its regulatory influence is crucial yet less pronounced in comparison to other constructs. These insights provide a strategic roadmap for enterprises to prioritize resource allocation and enhance their green

Table 7
CB-SEM dual test.

Index	Model fit	Reference	Result
ChiSqr/df	1.734	<3	Good
RMSEA	0.044	<0.08	Good
GFI	0.917	>0.90	Good
AGFI	0.899	>0.90	Acceptable
PGFI	0.755	>0.50	Good
SRMR	0.097	<0.08	Acceptable
NFI	0.927	>0.90	Good
TLI	0.964	>0.90	Good
CFI	0.968	>0.90	Good

innovation capabilities, especially by leveraging GenAI and supply chain innovations.

6. Study 2: fsQCA results

To provide a more nuanced understanding of the complex interplay between GenAI capability, AI regulation, and supply chain ambidexterity in driving internationalization green innovation performance, this study complements the PLS-SEM findings with a fsQCA. While PLS-SEM offers valuable insights into linear and symmetrical relationships, fsQCA excels in examining asymmetrical causal configurations, allowing for the identification of multiple pathways leading to the outcome of interest (Wang & Esperança, 2023). This combined approach provides a more comprehensive and robust understanding of the phenomenon under investigation. The initial step in fsQCA involves calibrating the causal conditions and outcome, transforming the Likert scale values of the variables into fuzzy membership scores (Wang & Esperança, 2023). This transformation reflects the degree to which each case belongs to a specific set, with scores ranging from 0.05 (full non-membership) to 0.95 (full membership), and 0.5 representing the crossover point. A necessity analysis revealed that no single construct exhibited a consistency score above 0.9, indicating the absence of a single necessary condition for achieving high levels of internationalization green innovation performance. This finding underscores the complex and multifaceted nature of the phenomenon, suggesting that multiple pathways and combinations of factors can contribute to success. Subsequently, a truth table was constructed to represent all possible combinations of causal conditions, with configurations selected for further analysis based on their frequency and consistency values (Wang & Esperança, 2023). This study set the minimum frequency to 3 cases, the raw consistency threshold to 0.80, and the proportional reduction in inconsistency (PRI) consistency to 0.6, ensuring a robust and reliable analysis. The fsQCA analysis yielded three solutions: complex, parsimonious, and intermediate solutions. This study opted for the intermediate solution for interpretation, striking a balance between completeness and parsimony while maintaining clarity and explanatory power (Table 8).

7. Study 3: interview results

To gain deeper insights into the quantitative findings and explore the nuanced experiences of firms implementing GenAI for internationalization green innovation, semi-structured interviews were conducted with 10 senior executives from cross-border e-commerce SMEs in China (Jayasena et al., 2024). The interviewees were purposefully selected based on their firms' active involvement in GenAI implementation and international market expansion. Each interview lasted approximately 45–60 min and was conducted in Mandarin Chinese via video conferencing. The interviews were audio-recorded with consent, transcribed verbatim, and translated into English for analysis (Table 9). Prior to the main interview phase, a pilot study was conducted with two senior executives from SMEs similar to those in our main sample but not included in the final interview sample. The purpose of the pilot study was to refine the interview protocol, test the clarity and relevance of the interview questions, and assess the interview duration. The pilot interviews were conducted following the same procedure as the main interviews. Feedback from the pilot participants was used to revise the interview protocol, including clarifying ambiguous questions, adding probing questions for greater depth, and adjusting the order of questions for better flow. The pilot study confirmed the feasibility of the interview procedure and the appropriateness of the 45–60 min duration. The interview protocol was designed to explore key themes emerging from the quantitative results, including: (1) motivations for adopting GenAI, (2) perceived impacts on supply chain innovation and green performance, (3) challenges in implementation and regulatory compliance, and (4) future plans for AI-driven sustainability initiatives. Thematic analysis was employed to identify recurring patterns and insights across the interview data (Jayasena et al., 2024).

Table 8
High levels of internationalization green innovation performance.

Construct	Configurations		
	1	2	3
GenAIC		◎	◉
AIR		◉	◉
SCII	◉	◉	
SCRI	◉		◉
Raw coverage	0.639	0.366	0.435
Unique coverage	0.157	0.038	0.037
Consistency	0.837	0.852	0.876
Solution coverage	0.714		
Solution consistency	0.809		

Note: '◉' indicates core conditions, while '◎' indicates their absence. Blank cells indicate 'irrelevant'.

The interviews revealed several key themes that corroborate and expand upon the quantitative findings. Regarding the motivations for adopting GenAI, interviewees consistently emphasized the technology's potential to enhance competitiveness in international markets. As I3 noted, "GenAI allows us to rapidly adapt our products and marketing strategies to diverse global consumer preferences, giving us an edge over competitors still relying on traditional methods." This aligns with the study's findings on the positive relationship between GenAI capability and internationalization green innovation performance. The perceived impacts on supply chain innovation were particularly salient, with interviewees highlighting both exploratory and exploitative benefits. I7 explained, "We've used GenAI to simulate and test novel supply chain configurations, leading to breakthroughs in sustainability and efficiency." Simultaneously, I4 described how "GenAI has allowed us to optimize existing processes, reducing waste and improving responsiveness to market demands." These insights support the mediating roles of supply chain exploratory and exploitative innovation identified in the

Table 9
Characteristics of the interviewees.

ID	Position	Enterprise size	Enterprise age	International market focus
I1	CEO	180	7	North America, Europe
I2	CTO	95	4	Southeast Asia, Middle East
I3	COO	275	11	Europe, Australia
I4	VP of Innovation	150	6	Japan, South Korea
I5	Director of Sustainability	210	9	Europe, South America
I6	Head of AI	120	5	North America, Southeast Asia
I7	Supply Chain Manager	230	8	Europe, Africa
I8	Chief Strategy Officer	185	7	Australia, New Zealand
I9	Director of IB	160	6	Europe, Middle East
I10	Head of R&D	140	5	Japan, Southeast Asia

quantitative analysis. Interestingly, the interviews revealed nuanced perspectives on AI regulation that add depth to the quantitative findings. While some interviewees viewed regulations as potential constraints, others saw them as catalysts for responsible innovation. I5 remarked, "Initially, we saw AI regulations as obstacles. However, they've actually pushed us to develop more robust and ethical AI systems, which has improved our reputation in environmentally conscious markets." This suggests that the relationship between AI regulation and innovation outcomes may be more complex than initially hypothesized. Challenges in implementation were frequently cited, with data quality and integration emerging as key concerns. I2 noted, "Ensuring the accuracy and relevance of training data for our GenAI models has been a significant hurdle, particularly when dealing with diverse international markets." This highlights the need for future research on data management strategies in the context of AI-driven internationalization. Looking ahead, interviewees expressed optimism about the future of AI-driven sustainability initiatives. I10 envisioned, "We're exploring how GenAI can help us design products with circular economy principles from the outset, potentially revolutionizing our approach to sustainability." This forward-looking perspective underscores the long-term potential of GenAI in driving green innovation, aligning with the study's broader implications.

These qualitative insights not only validate the quantitative findings but also provide rich contextual understanding of how SMEs are navigating the complexities of GenAI adoption for internationalization and sustainability. The interviews highlight the dynamic interplay between technological capabilities, regulatory environments, and innovative strategies, offering valuable directions for future research and practice in this rapidly evolving field.

8. Discussion

8.1. Summary of key findings and their relation to prior work

Our analysis reveals that GenAI capability exerts a significant, albeit modest, direct influence on internationalization green innovation performance ($\beta = 0.100$, $p < 0.05$, $f^2 = 0.012$). This finding aligns with emerging literature highlighting AI's potential to enhance environmental performance (Bolón-Canedo et al., 2024; Kanungo et al., 2024) but also suggests that the societal impact of GenAI is more nuanced than often assumed. It builds upon studies emphasizing the importance of contextual factors in technology adoption (Al-Emran & Griffy-Brown, 2023) and underscores the need for a more granular understanding of how GenAI specifically contributes to green innovation outcomes in internationalizing SMEs. We have added a more comprehensive comparison of our findings with recent studies that explore AI's role in sustainability, highlighting both convergence and divergence in results.

The strong positive relationships observed between GenAI capability and both supply chain exploratory innovation ($\beta = 0.350$, $p < 0.001$, $f^2 = 0.168$) and exploitative innovation ($\beta = 0.390$, $p < 0.001$, $f^2 = 0.213$) resonate with research demonstrating AI's capacity to enhance organizational ambidexterity (Wael Al-khatib et al., 2024). However, our study extends this understanding by specifically focusing on the supply chain context and differentiating between exploratory and exploitative innovation. We find that GenAI's impact on exploitative innovation is slightly stronger, suggesting that firms initially leverage GenAI to optimize existing supply chain processes rather than pursuing radical transformations (Wang & Zhang, 2025). This aligns with the resource-based view's perspective on incremental organizational change (Alkaraan et al., 2024) and provides empirical evidence for the theoretical proposition that firms often prioritize efficiency gains before embarking on more radical innovations.

8.2. Addressing study objectives and knowledge advancements

Our study aimed to investigate the mediating roles of supply chain exploratory and exploitative innovation in the relationship between GenAI capability and internationalization green innovation performance within the context of cross-border e-commerce SMEs. The confirmed mediating effects of both exploratory and exploitative supply chain innovation (supporting H6 and H7) directly address this objective. This represents a significant advancement in understanding the operational pathways through which GenAI influences green innovation outcomes. While prior research has acknowledged AI's potential to drive supply chain improvements (Fosso Wamba et al., 2024), our study provides empirical evidence of the specific mediating mechanisms, demonstrating how GenAI capabilities translate into enhanced green innovation performance through their impact on supply chain innovation.

Our study acknowledges the dynamic nature of GenAI capabilities, drawing upon the dynamic capabilities framework (Teece et al., 1997). We advance beyond a static interpretation of the Resource-Based View (RBV) by offering a more nuanced understanding of how these evolving capabilities contribute to sustained competitive advantage in a rapidly changing environment. By clearly differentiating between resources and capabilities and elucidating their interplay within the specific context of supply chain innovation, our research provides a refined and empirically testable application of the RBV. This addresses a key limitation highlighted by Hitt et al. (2016b), namely the need for a more dynamic perspective, thereby strengthening the theoretical foundation for understanding how GenAI can be leveraged for sustained competitive advantage. Furthermore, we incorporate the role of external relationships and networks, acknowledging the limitations of a purely internal RBV focus. This allows for a more holistic understanding of how firms can strategically leverage both internal and external resources to enhance internationalization green innovation performance. By situating our study within the specific context of Chinese cross-border e-commerce SMEs and acknowledging the influence of the regulatory environment, we address the limitation of RBV in neglecting contextual factors, as emphasized by recent research (Tiwari et al., 2024). This contextualized approach significantly enhances the relevance and applicability of our findings to both practitioners and scholars.

Our findings resonate strongly with the recent work of Shore et al. (2024), who empirically demonstrated that GenAI significantly bolsters entrepreneurial resilience in SMEs. Similar to their findings, our results underscore the critical role of GenAI in enhancing adaptability and resilience, particularly for SMEs navigating the complexities of dynamic international markets. Moreover, our findings align with the theoretical framework proposed by Dubey et al. (2024), which underscores the use of GenAI in benchmarking and improving supply chain management practices. Our results provide empirical evidence of how GenAI can be utilized to enhance both supply chain exploratory and exploitative innovation, ultimately contributing to a more robust and adaptable supply chain, as posited by their framework. Furthermore, our research extends the work of Li, Song, et al. (2024), who illuminated the mediating roles of green supply chain collaboration and circular economy implementation in the relationship between generative AI usage and sustainable supply chain performance. While Li, Song, et al. (2024) examined the broader context of sustainable supply chain management, our research offers a more focused and nuanced understanding of how these practices operate within the specific context of internationalizing SMEs. Additionally, our findings contribute to the emerging discourse on supply chain survivability, a concept brought to the forefront by Sarkar et al. (2025). Our study demonstrates how GenAI, through its impact on supply chain innovation, enhances not only resilience but also long-term viability, thereby contributing to the overall survivability of supply chains in the face of increasing disruptions. Finally, our findings complement the work of Lu et al. (2025), who highlight the importance of resource orchestration, encompassing internal reconfiguration and external integration, in enhancing supply chain resilience. We build upon their work by demonstrating empirically how GenAI capabilities facilitate this resource orchestration, enabling firms to adapt to

changing environments more effectively.

Furthermore, our investigation of AI regulation as a moderator contributes to the nascent literature on AI governance (De Almeida et al., 2021). The unexpected finding that AI regulation positively moderates the relationship between GenAI capability and supply chain innovation challenges the conventional view of regulation as a barrier to innovation. This represents a novel contribution to the field, suggesting that well-designed regulations can actually foster more responsible and sustainable technological practices by encouraging firms to develop more robust and ethical AI systems. This aligns with recent arguments about "responsible AI" driving both social and economic value (Kanungo et al., 2024) and expands our understanding of how regulatory frameworks can shape the societal impact of AI.

8.3. Connecting findings to technology in society's scope

Our study demonstrates that the impact of GenAI on sustainable development is not deterministic but mediated by complex social and organizational factors. The mediating role of supply chain innovation highlights how technological decisions ripple through organizational systems to shape broader societal and environmental impacts. Moreover, the positive moderating effect of AI regulation underscores the importance of societal governance mechanisms in guiding technological development toward beneficial outcomes. This highlights the agency that societies have in shaping the trajectory of AI adoption and its consequences. By illuminating these intricate relationships, our study contributes to the journal's mission of understanding how technology and society co-evolve and how technological choices can be intentionally directed toward positive social transformation.

9. Conclusions, implications, limitations, and future directions

9.1. Summary of findings and contribution to technology in society

Our findings illuminate the complex societal implications of GenAI adoption, demonstrating that its impact on sustainable development is mediated by organizational practices and regulatory frameworks. The analysis reveals that GenAI capability has a significant but modest direct effect on internationalization green innovation performance ($\beta = 0.100$, $p < 0.05$, $f^2 = 0.012$). This underscores the notion that while GenAI presents opportunities for enhancing environmental performance, its societal impact is more nuanced than often assumed and must be considered within its social and environmental contexts.

The strong positive relationships between GenAI capability and both supply chain exploratory innovation ($\beta = 0.350$, $p < 0.001$, $f^2 = 0.168$) and exploitative innovation ($\beta = 0.390$, $p < 0.001$, $f^2 = 0.213$) highlight how technological choices influence organizational transformation. The mediating effect of supply chain innovation in translating GenAI capabilities into sustainable outcomes further demonstrates how technological decisions ripple through organizational systems to shape broader societal and environmental impacts.

Perhaps most revealing for Technology in Society's focus on technology governance is our unexpected finding regarding AI regulation's positive moderating effect. Rather than constraining innovation, well-designed regulations appear to foster more responsible and sustainable technological practices, challenging deterministic views of technology-society relationships. The fsQCA results further reveal multiple pathways through which society can engage with GenAI for sustainable development, emphasizing the agency of communities and organizations in adopting and adapting AI technologies to serve their specific contexts and needs.

These findings contribute significantly to understanding how technological decisions shape social and environmental transformation, a core focus of Technology in Society. They demonstrate that the impact of GenAI on sustainable development is mediated by complex social and organizational factors, requiring thoughtful consideration of how societies can intentionally engage with these technologies to promote positive transformation.

9.2. Theoretical implications

This study makes several significant contributions to understanding how technological decisions shape societal transformation through the lens of GenAI adoption. First, it extends societal transformation theory by demonstrating how GenAI capability, as a transformative technological force, influences the social and environmental dynamics of international business (Dabić et al., 2023; Kumar et al., 2024; Wang & Esperança, 2023). This finding illuminates how technological choices at the firm level can cascade into broader societal changes, particularly in driving sustainable practices across global supply chains. Second, the study advances our understanding of how technological decisions intersect with organizational social structures by revealing the mediating role of supply chain innovation. This demonstrates how technological choices are embedded within and shaped by existing social systems and organizational practices, particularly in the context of sustainability transitions.

The study makes a crucial contribution to understanding the social forces that shape technological decisions by examining the role of AI regulation in moderating innovation outcomes. Rather than viewing regulations as mere technical constraints, our findings reveal them as important social mechanisms that guide technological development toward societal benefit (De Almeida et al., 2021). This challenges traditional assumptions about regulation hampering innovation and suggests that societal governance frameworks can actually foster more responsible and sustainable technological development (Díaz-Rodríguez et al., 2023).

Our methodological integration of PLS-SEM, IPMA, and fsQCA offers a comprehensive framework for analyzing how technological decisions are shaped by and in turn shape social forces. This approach reveals the complex interplay between technological capabilities, organizational practices, and regulatory environments in driving societal transformation toward sustainability. The findings highlight how different configurations of technological and social factors can lead to similar sustainability outcomes, emphasizing the importance of considering local contexts and social dynamics in technological decision-making.

These theoretical contributions provide a foundation for future research on how technological choices in the age of GenAI can be intentionally directed toward positive societal transformation. Future studies could explore how different stakeholder groups influence technological decisions, how cultural contexts shape AI adoption patterns, and how technological choices in one society affect sustainable development in others.

9.3. Practical implications

This study provides crucial insights for decision-makers seeking to thoughtfully engage with GenAI in ways that promote positive societal transformation. First, our findings emphasize that technological decisions about GenAI implementation should be considered within broader societal contexts. While the direct impact on green innovation may be modest, GenAI's influence on organizational social structures and practices suggests its potential as a catalyst for broader societal change (Lin et al., 2022; Wael Al-khatib et al., 2024).

Decision-makers should approach GenAI adoption not merely as a technical challenge but as a socio-technical transformation requiring careful consideration of human capital, organizational culture, and societal implications (Chen et al., 2022; Mikalef & Gupta, 2021; Neiroukh et al., 2024). This includes fostering organizational cultures that balance technological efficiency with social responsibility and environmental sustainability.

The study highlights the importance of proactive engagement with societal governance mechanisms. Rather than viewing AI regulations as

constraints, decision-makers should recognize them as social frameworks that can guide responsible innovation (Li, Song, et al., 2024). Active participation in policy dialogues and multi-stakeholder initiatives can help shape technological development that serves broader societal goals (Chowdhury et al., 2024).

By adopting this socially-conscious approach to technological decision-making, organizations can contribute to intentional and positive societal transformation while pursuing their business objectives (Al-Emran & Griffy-Brown, 2023). This includes considering how technological choices affect different stakeholder groups, environmental sustainability, and social equity in global markets.

9.4. Policy implications for societal transformation

Our findings offer crucial insights for policymakers seeking to shape the societal impact of GenAI through responsible innovation and sustainable development (Coeckelbergh & Sætra, 2023). The study reveals how technological decisions around GenAI adoption can transform social and economic dynamics, particularly for SMEs. First, our results highlight the need for policies that democratize access to GenAI capabilities across society. Beyond mere funding and technical support, policymakers must consider how GenAI access shapes power dynamics and social equity in the digital economy (Rajaram & Tinguely, 2024). This requires developing inclusive support programs that consider both technological and social barriers to AI adoption. The concept of supply chain survivability becomes particularly relevant here, emphasizing the need for supply chains to not only recover from disruptions but also to adapt and thrive in the long term (Sarkar et al., 2025).

The study demonstrates how AI regulation can actively shape technological choices and their societal outcomes. Rather than viewing regulations as purely technical constraints, policymakers should frame them as tools for guiding technology's role in social transformation (De Almeida et al., 2021; Díaz-Rodríguez et al., 2023). This perspective suggests developing participatory approaches to AI governance that engage diverse stakeholders in determining how AI technologies should serve society's needs. Such frameworks must balance innovation with ethical considerations while addressing broader societal concerns about AI's impact on employment, privacy, and social cohesion.

Our findings emphasize the need for policies that foster global technological cooperation while respecting local social contexts. As environmental challenges transcend borders, policies must facilitate knowledge sharing while remaining sensitive to different societies' values and needs (Saheb and Saheb, 2023). This calls for creating collaborative platforms that enable cross-cultural dialogue about AI's role in addressing sustainability challenges. By thoughtfully engaging with these policy dimensions, decision-makers can help ensure that GenAI's technological capabilities advance societal well-being while promoting sustainable development (Jayasena et al., 2024).

9.5. Limitations and future directions

This study, while providing valuable insights, has several limitations that warrant consideration from a technology-in-society perspective. First, our focus on Chinese SMEs limits our understanding of how different social, cultural, and institutional contexts might influence GenAI's transformative potential. Future research should examine how varying societal values and structures shape technological decisions around AI adoption. Second, our primarily organizational focus may understate the broader social forces that influence technological decision-making, including public attitudes, cultural norms, and political dynamics.

These limitations suggest several promising directions for technology-in-society research. First, investigating how GenAI reshapes social relationships and power dynamics throughout supply chains could illuminate the broader societal implications of technological change. Second, examining how different stakeholder groups influence and are affected by AI-related technological decisions would provide valuable insights for more inclusive innovation governance. Future research should explore how resource orchestration strategies, as highlighted by Lu et al. (2025), vary across different industries and contexts, and how these strategies interact with AI assimilation to impact supply chain resilience. Future research should examine how varying societal values and structures shape technological decisions around AI adoption in different settings, considering factors such as regulatory frameworks, industry characteristics, and cultural norms (Tiwari et al., 2024). By pursuing these research directions, scholars can better understand how technological choices shape - and are shaped by - the ongoing transformation of society. This knowledge is crucial for decision-makers seeking to thoughtfully engage with technology's role in creating a more sustainable and equitable world.

CRediT authorship contribution statement

Shaofeng Wang: Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization. **Hao Zhang:** Writing – review & editing, Writing – original draft.

Ethics approval

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Declaration of competing interest

The authors have no conflicts of interest to declare.

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Data availability

Data will be made available on request.

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